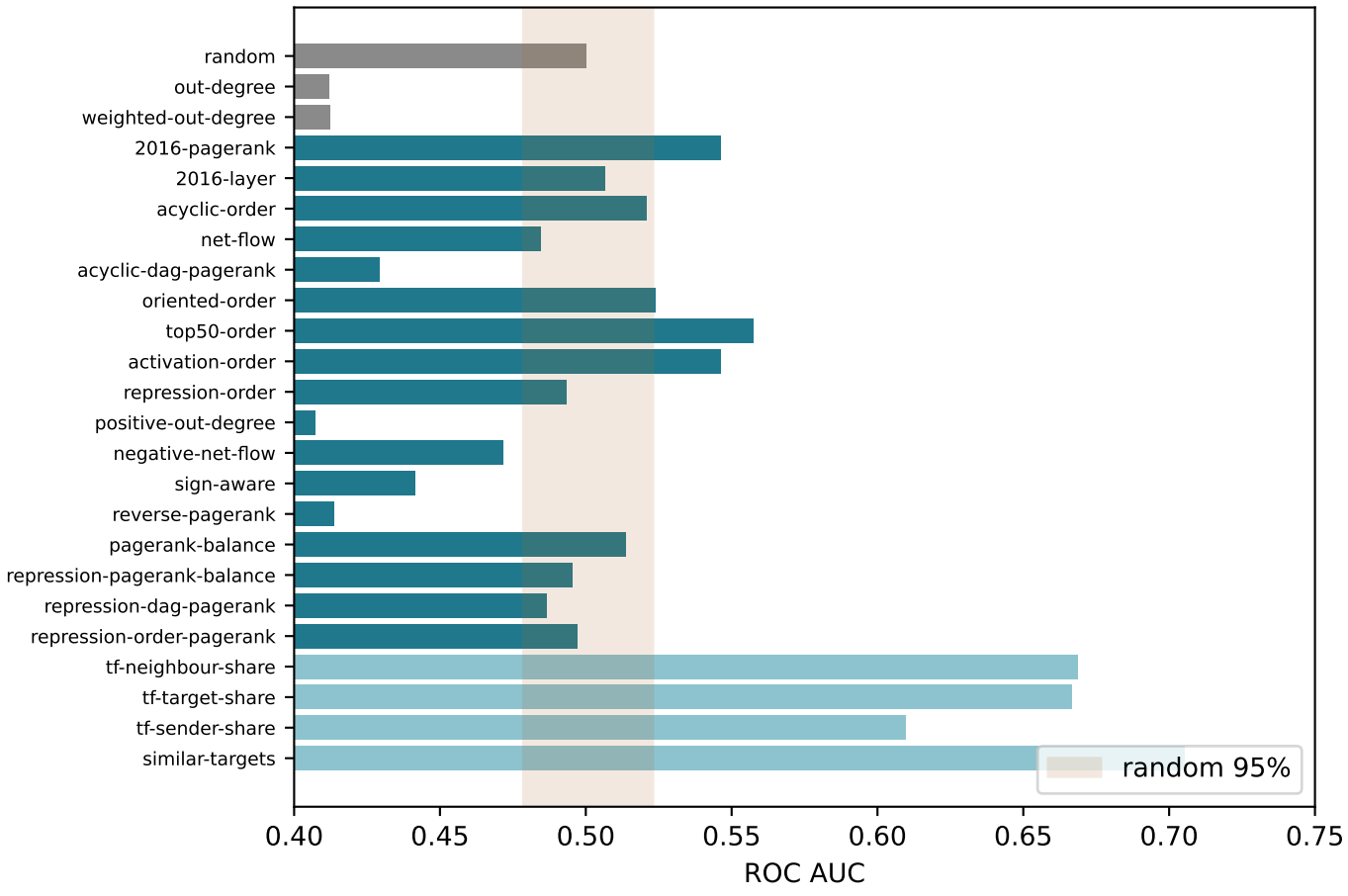
# Regulator ranking against reference 'lambert-2018'

2026-11-10 · 8,708 genes ranked, 752 in lambert-2018-tfs.txt · top k = 100

| Pipeline | Reference genes in top 100 | AUC | p top-k | p AUC | AUC vs best baseline (95%) | p vs baseline |
|---|---|---|---|---|---|---|
| random | 8.7 (4–14) | 0.500 (0.479–0.523) | | | | |
| out-degree | 8 | 0.412 | 0.641 | 1.000 | -0.000 (-0.002, +0.001) | 0.614 |
| weighted-out-degree | 8 | 0.412 | 0.641 | 1.000 | +0.000 (+0.000, +0.000) | |
| 2016-pagerank | 14 | 0.546 | 0.052 | < 0.001 | +0.134 (+0.105, +0.164) | < 0.001 |
| 2016-layer | 6 | 0.507 | 0.881 | 0.274 | +0.094 (+0.071, +0.119) | < 0.001 |
| acyclic-order | 6 | 0.521 | 0.881 | 0.039 | +0.109 (+0.079, +0.137) | < 0.001 |
| net-flow | 8 | 0.485 | 0.641 | 0.928 | +0.072 (+0.052, +0.092) | < 0.001 |
| acyclic-dag-pagerank | 7 | 0.429 | 0.771 | 1.000 | +0.017 (+0.008, +0.026) | < 0.001 |
| oriented-order | 12 | 0.524 | 0.160 | 0.020 | +0.112 (+0.082, +0.141) | < 0.001 |
| top50-order | 5 | 0.558 | 0.945 | < 0.001 | +0.145 (+0.112, +0.180) | < 0.001 |
| activation-order | 5 | 0.546 | 0.945 | < 0.001 | +0.134 (+0.102, +0.168) | < 0.001 |
| repression-order | 6 | 0.494 | 0.881 | 0.724 | +0.081 (+0.060, +0.101) | < 0.001 |
| positive-out-degree † | 6 | 0.407 | 0.881 | 1.000 | -0.005 (-0.011, +0.001) | 0.118 |
| negative-net-flow † | 8 | 0.472 | 0.641 | 0.995 | +0.060 (+0.041, +0.077) | < 0.001 |
| sign-aware † | 10 | 0.442 | 0.384 | 1.000 | +0.029 (+0.018, +0.041) | < 0.001 |
| reverse-pagerank † | 7 | 0.414 | 0.771 | 1.000 | +0.001 (-0.004, +0.007) | 0.556 |
| pagerank-balance † | 8 | 0.514 | 0.641 | 0.119 | +0.101 (+0.076, +0.126) | < 0.001 |
| repression-pagerank-balance † | 7 | 0.495 | 0.771 | 0.667 | +0.083 (+0.058, +0.106) | < 0.001 |
| repression-dag-pagerank † | 7 | 0.487 | 0.771 | 0.900 | +0.074 (+0.055, +0.092) | < 0.001 |
| repression-order-pagerank † | 7 | 0.497 | 0.771 | 0.598 | +0.085 (+0.065, +0.103) | < 0.001 |
| tf-neighbour-share ‡ | 58 | 0.669 | < 0.001 | < 0.001 | +0.256 (+0.229, +0.281) | < 0.001 |
| tf-target-share ‡ | 37 | 0.667 | < 0.001 | < 0.001 | +0.255 (+0.228, +0.283) | < 0.001 |
| tf-sender-share ‡ | 49 | 0.610 | < 0.001 | < 0.001 | +0.197 (+0.171, +0.222) | < 0.001 |
| similar-targets ‡ | 58 | 0.705 | < 0.001 | < 0.001 | +0.293 (+0.262, +0.323) | < 0.001 |

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



# Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

## collectri

| Graph | Pairs | Recall (×) | Positive × | Negative × | Sign (chance) |
|---|---|---|---|---|---|
| raw graph | 25,488 | 0.038 (1.75×) | 3.71× | 0.74× | 0.63 (0.65) |
| resolve_reciprocals → drop_sinks → prune_triangles | 25,353 | 0.008 (0.94×) | 4.72× | 0.87× | 0.16 (0.21) |
| resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering | 16,723 | 0.010 (0.76×) | 3.17× | 0.75× | 0.14 (0.14) |
| absolute | 25,488 | 0.038 (1.75×) | | | |
| absolute → minimum_feedback_order | 25,195 | 0.029 (1.46×) | | | |
| orient_reciprocals ratio=2.0 → absolute | 25,455 | 0.023 (1.27×) | | | |
| orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order | 25,455 | 0.023 (1.25×) | | | |
| top_outgoing k=50 → absolute | 25,488 | 0.012 (4.98×) | | | |
| top_outgoing k=50 → absolute → minimum_feedback_order | 25,411 | 0.009 (4.31×) | | | |
| keep_sign sign=1 → absolute | 24,598 | 0.028 (3.75×) | | | |
| keep_sign sign=1 → absolute → minimum_feedback_order | 24,417 | 0.020 (3.74×) | | | |
| keep_sign sign=-1 → absolute | 15,859 | 0.017 (0.65×) | | | |
| keep_sign sign=-1 → absolute → minimum_feedback_order | 15,570 | 0.017 (0.64×) | | | |
| keep_sign sign=-1 | 15,859 | 0.017 (0.65×) | | 0.65× | 0.11 (0.11) |

# Pipeline steps

### out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,277,515 edges (1,387,746 negative), largest strong component 8,645.

### weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,277,515 edges (1,387,746 negative), largest strong component 8,645.

### 2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 899,117 edges (880,327 negative), largest strong component 4,926.

### 2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 899,117 edges (880,327 negative), largest strong component 4,926; DAG keeps 704,363 edges.

### acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,277,515 edges (0 negative), largest strong component 8,645; DAG keeps 1,992,586 edges.

### net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,277,515 edges (0 negative), largest strong component 8,645.

### acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,277,515 edges (0 negative), largest strong component 8,645; DAG keeps 1,992,586 edges.

### oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,846,577 edges (0 negative), largest strong component 4,001; DAG keeps 1,831,741 edges.

### top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 291,418 edges (0 negative), largest strong component 5,997; DAG keeps 222,522 edges.

### activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 889,769 edges (0 negative), largest strong component 5,454; DAG keeps 630,122 edges.

### repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,387,746 edges (0 negative), largest strong component 1,485; DAG keeps 1,384,210 edges.

### positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,277,515 edges (1,387,746 negative), largest strong component 8,645.

### negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,277,515 edges (1,387,746 negative), largest strong component 8,645.

### sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,277,515 edges (1,387,746 negative), largest strong component 8,645.

### reverse-pagerank: Highest reverse PageRank (raw graph)